it also reduces the cooking time. Jellies are clear substances made of fruit juice or the extract of a fruit.

Sugar acts in the following ways:

- Sugar draws the water out of food therefore making it unavailable for microorganisms.
- As a result of water loss, microbial metabolism is stopped.
- Hence, the growth of microorganisms is stopped

### Preparation of jelly:

- Under-ripe fruits are used, because the pectin content is high and good acidity is essential for a good jelly.
- Pieces of fruit are completely immersed in water and cooked for 10–20 minutes. Hard fruits like guavas need to be cooked for 45 minutes.
- After the fruit is cooked, it is strained without disturbing the fruit pieces.
- The fruit extracts contain pectin which determines the addition of sugar. When the level of pectin is high, it needs more sugar but requires less boiling time.
- Rapid boiling facilitates rapid evaporation, which avoids strong flavour and darkened colour.
- Then the jelly is poured in bottles or moulds, and allowed to set without any disturbance.

CASE STUDY

**Case study 1**

After a grand marriage function, 50 bananas, 5 kgs of tomatoes and 3 kgs of lime were left unused. Suggest ways to preserve these food items.

### Preparation of jam:

- Fruits like apples are cooked with skin and made into pulp with the strainer for making jam
- Equal quantities of sugar and pulp are taken to make jam.
- After it is cooked, it is transferred to a sterilized bottle and allowed to cool.

### Test for doneness for jam

Sheet test – the mixture is allowed to drip from a large cool spoon. If the syrup forms a sheet instead of two separate drops, the jam is done.

Bubble test – when the end point reaches, big bubbles can be seen throughout the jam.

Plate test – set a plate in the freezer for some time. Put the jam and tilt the plate slowly. The jam should come down as a whole mass forming "U" shape. Water should not separate out.

Fork test – dip the fork into the jam or jelly. Jam of correct consistency forms a sheet between the needles of the fork.

### Honey

Honey is a natural preservative in its original state and was one of the earliest preservatives used by ancient civilizations. It has a high concentration of sugar that draws out the water out of yeast or bacteria cells which contaminate the food.

## 4.2.6.2 High Concentration of Salt

Foods are also preserved by the principle of osmotic pressure in salting and pickling. Most commonly used preservative is sodium chloride. Required quantity may be added to slow down or prevent the growth of microorganisms or enough to permit lactic acid fermentation to take place.



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